

**DR.BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE-RAIGAD 402103
End Semester Examination**

Department: Electronics and Telecommunication Engineering
Subject with Subject Code: Analog Circuits (BTEXC302)
Date: 29-05-2019

Semester: III
Marks: 60
Time: 3 Hrs

Instructions to Students:

***Each question carries equal marks.**

***Assume suitable data wherever necessary and mention it clearly**

***All the notations mentioned in this question paper have their usual meaning.**

Q.1 A What is an Op-amp ? Draw its block diagram and explain the function of each block in detail. [5]

Q.1 B The following specifications are given for dual input balanced-output differential amplifier $R_C = 2.2K\Omega$, $R_B = 4.7K\Omega$, $R_{i1} = R_{i2} = 50\Omega$, $V_{CC} = 10V$, $-V_{EE} = -10V$, $\beta_{dc} = 100$ and $V_{BE} = 0.715V$. Determine the voltage gain, input resistance and output resistance. [5]

Q.2 SOLVE ANY TWO QUESTIONS.

Q.2 A Explain the configuration of inverting amplifier and derive the expression for its gain. [5]

Q.2 B Draw the circuit of Op-amp as a difference amplifier and derive the expression for its output voltage. [5]

Q.2 C Draw the circuit of Op-amp as logarithmic amplifier and an exponential amplifier and derive the expression for their output voltage. [5]

Q.3 SOLVE ANY TWO QUESTIONS.

Q.3 A Explain any five characteristics of practical Op-amp in brief. [5]

Q.3 B Explain Half wave precision rectifier with neat diagram. [5]

Q.3 C Explain Schmitt Trigger with neat diagram & waveform. [5]

Q.4 A With neat diagram describe R-2R Ladder DAC. [5]

- Q.4 B What is the maximum conversion time for an 8-bit, 12-bit and 16-bit counting ADC when the clock frequency is 1 Mhz? [3]
- Q.4 C Explain V to I Converter. [2]
- Q.5 A A Colpitts Oscillator circuit having two capacitors of 24nF and 240nF respectively are connected in parallel with an inductor of 10mH. Determine the frequency of oscillations of the circuit, the feedback fraction and draw the circuit. [5]
- Q.5 B Draw a neat diagram of Wien Bridge Oscillator using IC741 and explain it. [5]
- Q.6 A Briefly explain role of Low Pass Filter and Voltage Controlled Oscillator in PLL. [5]
- Q.6 B List applications of PLL and briefly describe the role of the PLL in Frequency multiplier. [5]

-----All the Best-----